TECHNICAL ROADMAP TEMPLATE

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TEAM INFORMATION

Team Name	CardinalMu ASEAN
Institution	Mapúa University
Country	Philippines
Track	[✓] Climate Change [ ] Telemedicine [ ] Smart Cities [ ] AI for Education
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SECTION 1: EXECUTIVE SUMMARY (PROBLEM-SOLUTION FIT)

In December 2019, Typhoon Kammuri made landfall in the Philippines and moved westward across the South China Sea. Three days later, it generated catastrophic flooding across coastal Vietnam, but official Vietnamese government advisories arrived only after the storm was already bearing down on populated coastlines. The families in Quang Ngai province who lost their homes had smartphones, Facebook accounts, and an active internet connection. What they lacked was a system that could observe what was happening in the Philippines and, in Vietnam, tell them through the platforms they already used daily that it was coming for them. SEABeacon is that system.

SEABeacon solves this by bridging the gap in the Philippines-Vietnam hazard corridor and beyond. We aggregate official meteorological data, free RSS news feeds, and public Telegram channels, combining physical models with social signal layers, using an NLP classifier to detect distress signals in Filipino and English across public feeds. Rather than issuing legally risky evacuation orders, SEABeacon generates "Aggregated Advisories" localized into primary regional languages (e.g., Filipino, Vietnamese, and English), with secondary dialect support for Cebuano, extensible to minority languages such as Khmer in future phases. To ensure it reaches the most at-risk populations, like a rural fisherman in Eastern Samar, the system prioritizes SMS and Cell Broadcast Services alongside social platforms. SEABeacon transforms delayed diplomatic notifications into actionable, localized intelligence, designed to scale across all ten ASEAN member states as a modular, open architecture.

SECTION 2: TECHNICAL ARCHITECTURE

2.1 Scope and Boundaries:
In-Scope (Phase 1 MVP):

- **Hydro-Meteorological Hazards:** Typhoons, Cyclones, and Severe Cross-Border Flooding.
- **Airborne Geological Hazards:** Volcanic ash dispersion and transboundary haze.

- **Tsunami Relay:** Distribution of Pacific Tsunami Warning Center (PTWC) alerts.

Out-of-Scope:

- **Instant-Impact Tectonic Events:** Earthquakes (due to sub-second sensor requirements).
- **Official Evacuation Orders:** SEABeacon strictly issues informational "Aggregated Advisories."

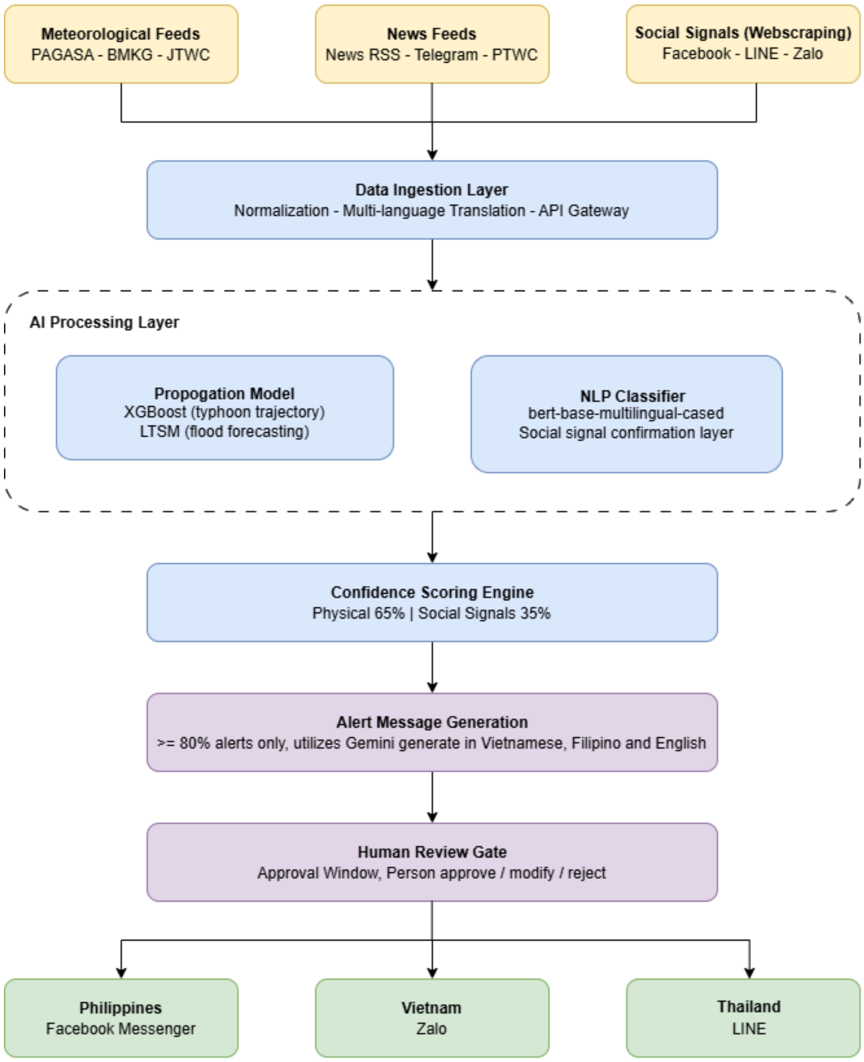
2.2 System Components:

Inputs: The system ingests public physical model data (historical typhoon tracks, pre-computed HYSPLIT volcanic ash scenarios) and social signals via RSS news feeds (Rappler, VnExpress) and public Telegram disaster channels (NDRRMC).

Processing Core: A cloud-based ML/NLP engine calculates a numeric confidence score by combining physical data and social signals. It runs a Three-Tier Alert Protocol (Watch, Advisory, Warning) to gate the alerts and prevent false positives.

Outputs: The system delivers two formats of localized Aggregated Advisories: a rich format for messaging apps (Messenger, LINE, Zalo) and an ultra-compressed 160-character format for SMS. Every alert includes mandatory legal buffer language directing users to official authorities.

2.3 AI Solution Diagram:



SECTION 3: AI APPROACH & MODEL SELECTION

Primary AI Approach: [X] Machine Learning [X] NLP [ ] Computer Vision [ ] Generative AI

- **Model Selection & Logic:**

Physical hazard propagation modeling (XGBoost for typhoon trajectory; LSTM for flood forecasting). Predicts where and when a hazard will arrive and at what severity. The project utilizes three AI approaches: Machine Learning (XGBoost, LSTM) for physical hazard modeling (typhoon/flood prediction); Natural Language Processing (bert-base-multilingual, opus-mt) for social signal detection and alert translation across six languages; and Generative AI (Jinja2) for deterministic, auditable alert text generation from structured data.

Scope / Components	Key Models & Frameworks	Primary Justification
Typhoon, Flood, Volcanic, Social, & Translation	XGBoost (Scikit), LSTM (Keras), BERT/Helsinki (HF), Jinja2	Ensures sub-100ms inference, temporal dependency capture, and deterministic, hallucination-free alerts.

- **Reasoning (The Confidence Protocol):**

A massive predictive generative model is a liability if it hallucinates a disaster. By utilizing a rigid Confidence Scoring System, we multiply the confidence of the physical model (max 65%) with the social signal (max 35%) to trigger alerts. This deterministic ML approach, combined with a mandatory 5-minute human review window for mass alerts, strictly prevents false positives while maintaining a Glass-to-Glass latency of under 3 minutes.

SECTION 4: DATA STRATEGY & ETHICS

4.1 Training Data Sources:

Dataset	Source	Hazard	Size/Coverage	License
Typhoon historical tracks	IBTrACS (NOAA)	Typhoon	~4,000 tracks (1851-2024)	Public domain
Disaster impact records	EM-DAT (CRED, UCLouvain)	All hazards	23,000 ASEAN events	Free non-commercial
Disaster event archive	GDACS (EU Joint Research Centre)	All hazards	20-year archive + live	Creative Commons
Mekong historical gauge data	Mekong River Commission	Flood	50-year gauge records	Free
NLP disaster corpus	Custom — RSS + Telegram (manually labeled)	All hazards	~5,000 labeled posts	Proprietary (team-curated)

4.2 Operational Data (Real-Time Feeds):

Source	Agency	Update Frequency	Access Method
Typhoon track and intensity	JTWC, JMA, PAGASA	Every 6 hours	Public HTTP / RSS feed
Rainfall forecasts	PAGASA, BMKG, NASA IMERG	Every 3 hours	Public API (open access)
Volcanic eruption alerts	PVMBG, PHIVOLCS	Event-driven	Public RSS feed

Ash cloud trajectories	VAAC Darwin (pre-computed HYSPLIT	On eruption event	ICAO NOTAM public feed
Tsunami advisories	Pacific Tsunami Warning Center (PTWC)	Event-driven	IANA EDXL syndication
Regional coordination	AHA Centre Open Data	As published	Open API
Social signals	Regional RSS news, Telegram public channels	Continuous polling	Public — no authentication required

4.3 Data Quality & Cleaning:

Different agencies use different wind speed measurement standards (1-minute vs. 10-minute sustained), coordinate reference systems, and hazard classification vocabularies. The ingestion layer normalizes all incoming data to WGS84 coordinates, 10-minute sustained wind speed (WMOSTANDARD), and a unified internal hazard taxonomy cross-indexed to GDACS event codes. Missing data points are handled using last-known-value carry-forward for up to 30 minutes. After 30 minutes without an update, the agency is flagged unreachable, and the pipeline falls back to the next available regional source. All normalization steps are logged with source attribution for auditing.

4.4 Licensing & Legality:

All data sources used in SEABeacon have been reviewed for license compatibility with hackathon and non-commercial academic use. The table below documents the license type, permitted use, and any conditions or restrictions applicable to each source.

Data Source Groups	License Type & Key Condition
Public Domain / Open Data	IBTrACS, JTWC, NASA IMERG. **Condition:** Freely usable; attribution to NASA recommended.
Attribution Required	GDACS, WorldPop, Smithsonian, Mekong River Commission. **Condition:** Creative Commons/Research use; requires citation.
Non-Commercial Use	EM-DAT, GADM. **Condition:** Limited to non-commercial academic/research use; registration required. (GADM cannot be redistributed standalone.)
Public Agency Feeds	PAGASA, BMKG, VNMHA, PVMBG, PHIVOLCS. **Condition:** Public RSS/HTTP; no raw data redistribution as a competing product.

4.5 Bias Mitigation & Fairness:

Language data bias

bert-base-multilingual performs significantly better on high-resource languages (Filipino, Bahasa Indonesia) than on low-resource languages (Burmese, Khmer). For MVP, the NLP classifier outputs a language-specific confidence flag. Alerts generated from low-resource language social signals are assigned a reduced social signal weight (20%, down from the standard 35%) in the confidence fusion formula, requiring stronger physical model confirmation before advancing to Advisory or Warning tier.

SECTION 5: DEVELOPMENT MILESTONES (AGILE ROADMAP)

Phase	Summary Activities	Tools	Outcome
Sprint 0: Concept	Lock project name, AI decisions, and alert framework; initialize repo.	GitHub, Google Docs	Architecture locked; repo live.
Sprint 1: Found.	Set up cloud/PostGIS; build ingestion scripts and LGU registries.	AWS/GCP, FastAPI, Python	Live data pipeline; cleaned datasets.
Sprint 2: AI	Train XGBoost/BERT models; implement confidence scoring and template engine.	scikit-learn, TensorFlow	Verified AI pipeline; Kammuri demo verified.
Sprint 3: Integ.	Build review dashboard; integrate Messenger/LINE/Telegram and SMS APIs.	React, Twilio, Messenger API	End-to-end prototype functional.
Sprint 4: Pitch	Final replay tests, stress-testing, video recording, and pitch deck finalization.	Loom, Canva, Figma	Final demo and pitch ready.

SECTION 6: SCALABILITY & REGIONAL RESILIENCE

6.1 Modular Country Architecture :

Each country in SEABeacon is a self-contained configuration module plugged into the shared core engine. A country module consists of four configuration layers: data feed configuration (which agency APIs to ingest and at what intervals), language configuration (which alert template language and which translation model), platform configuration (which messaging API and account credentials), and LGU registry (the contact endpoint database for all administrative regions). The core AI pipeline, the propagation model, the NLP classifier, and the confidence engine do not change across countries. Adding Indonesia requires configuring these four layers, plus additional data feeds (PVMBG volcanic alerts and Indonesian RSS). Estimated additional effort per country module: 3 to 5 days of configuration and integration testing.

6.2 Country Coverage & Roles:

Country	Phase	Hazard Role	Delivery
Philippines	MVP (P1)	Pilot: Typhoon origin/source	Messenger, Telegram, SMS (Primary demo)
Vietnam	P1	Cross-border typhoon recipient	Zalo Official Account (Kammuri demo target)
Thailand	P1	Haze, Mekong flood corridor	LINE Broadcast API

6.3 Technical Constraints & Mitigations :

Constraint	Impact	Mitigation
API verification delay, weaker NLP for low-resource languages, limited HYSPLIT computation, manual LGU registry, and a 5-minute human review gate.	Potential alert delivery delays, lower social signal recall, limited volcanic ash scenarios, risk of delivery failures, and mass-broadcast warnings delayed by up to 5 minutes.	Using Telegram for the hackathon demo, reducing social signal weight to 20%, pre-computing the 5 highest-risk scenarios, implementing a failure-flagging manual review (24h), and accepting the 5-minute delay as a safety trade-off.